Agentic AI Frameworks

✓ Autogen

CrewAI

GPTOSS 20b

User → msg1 → Writer LLM → msg2 Output (Article) → Reviewer LLM → Approved

msg3 (Reviewer LLM → Writer LLM)

- Problem Challenge ✓
- Solution ← Code, concepts
- Limitation ← X
- Conclusion ✓

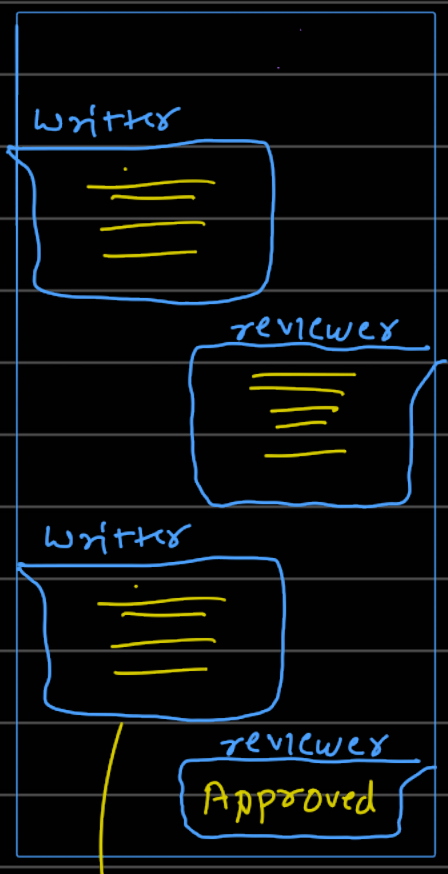
SDL general & runner

↓

reviewer → Approved

↓ Reject

Data collector



↓

Output

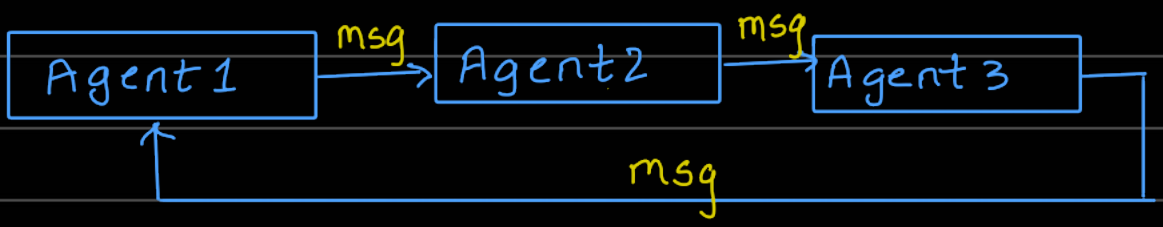
1) Assistant Agent → Writter, Reviewer

2) Messages → Information exchanged

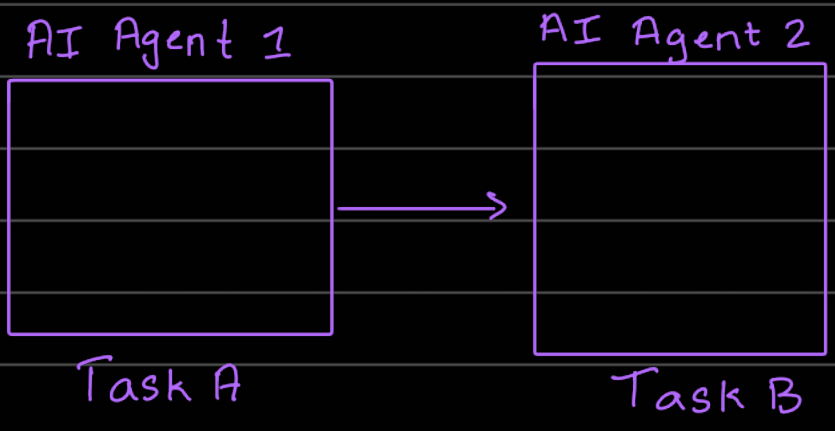
3) RoundRobinGroupChat → Fixed order

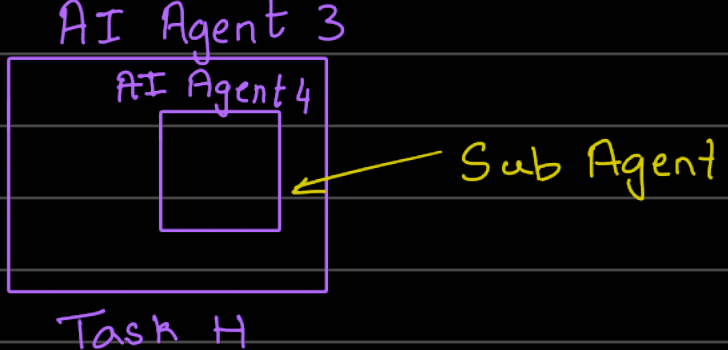
4) SelectorGroupChat → Selects next agent

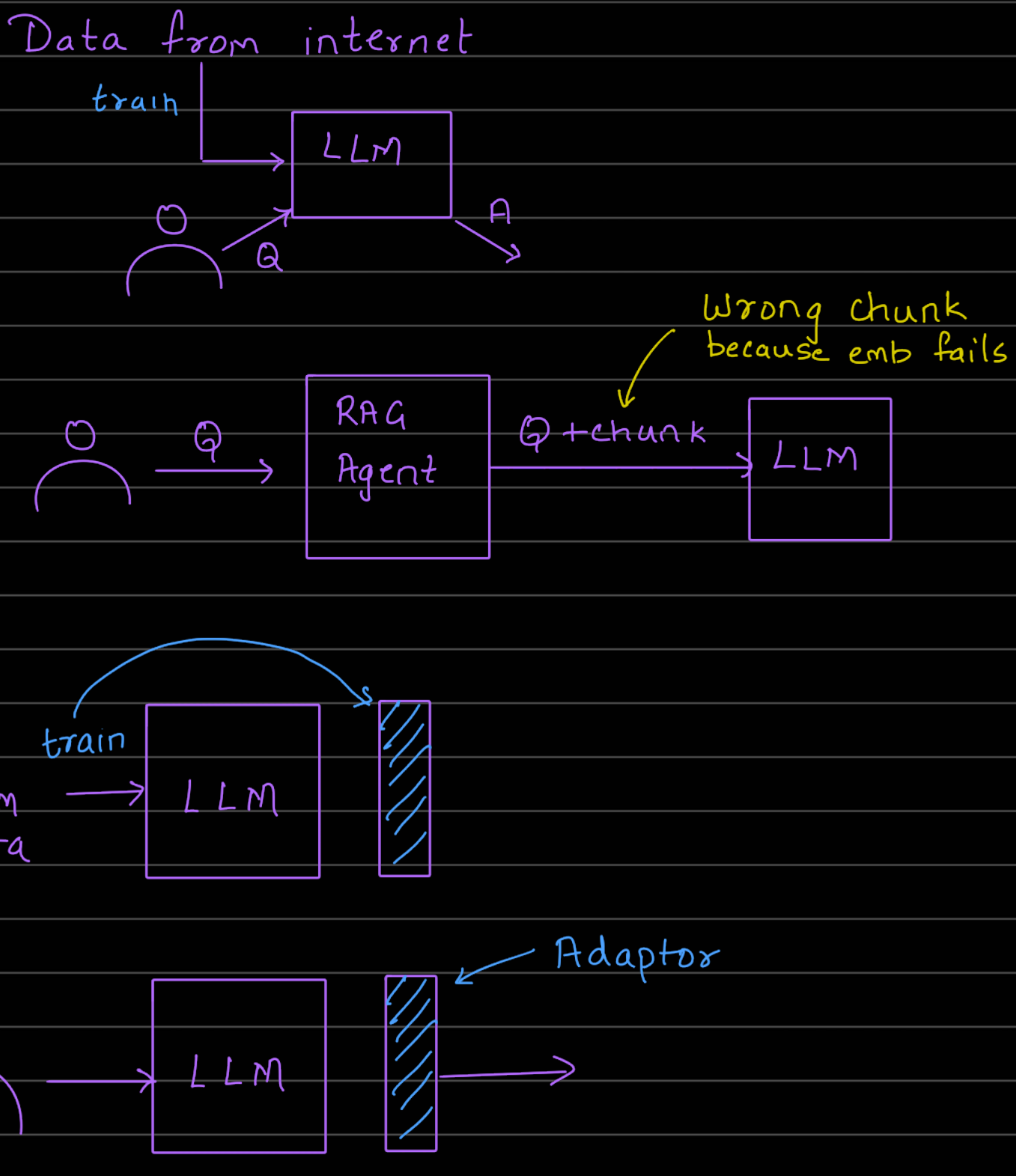
5] State → Stored information



Multi Agent







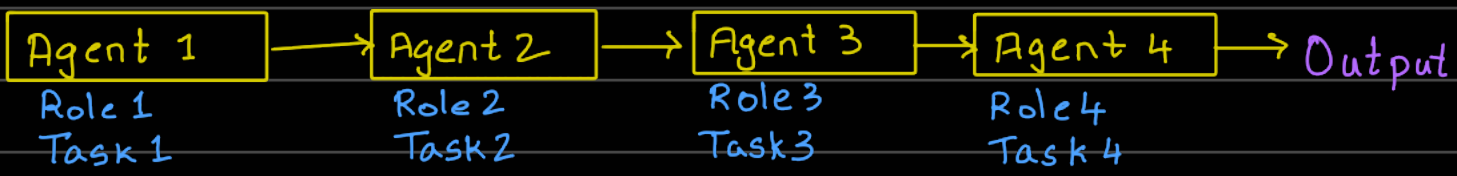
Adaptor is not directly trained
It is divided into 2 low rank matrix

Low Rank Adaptors = LoRA

CrewAI

Crew (Group of agents)

Agent → 1) Role
2) Task
3) Context
4) Model



Role = Writer
Task = Write article on Agentic AI
Model = GPT OSS 20b

↓

Role = Scriptor
Task = Make article engaging
Model = GPT OSS 20b

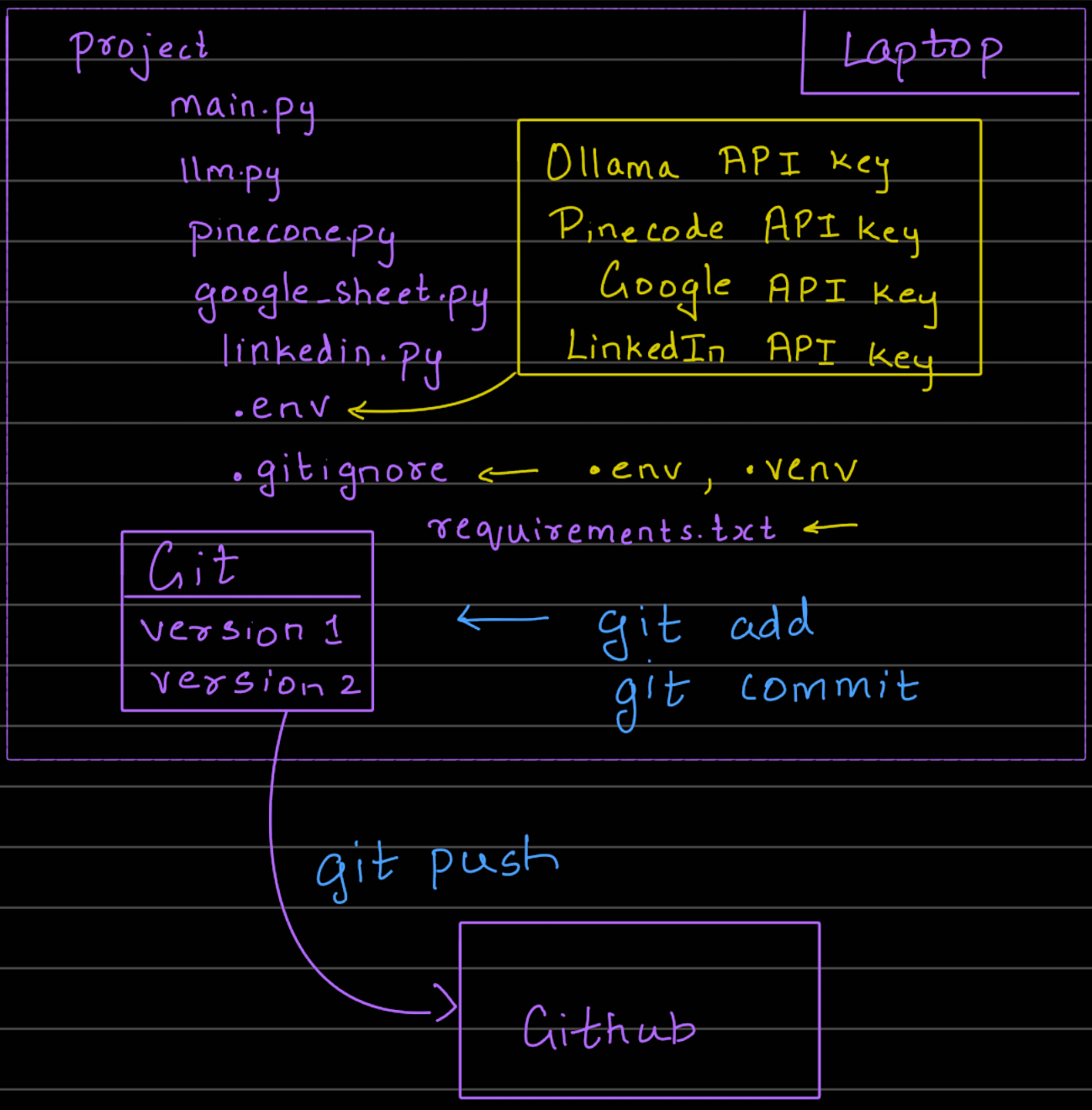
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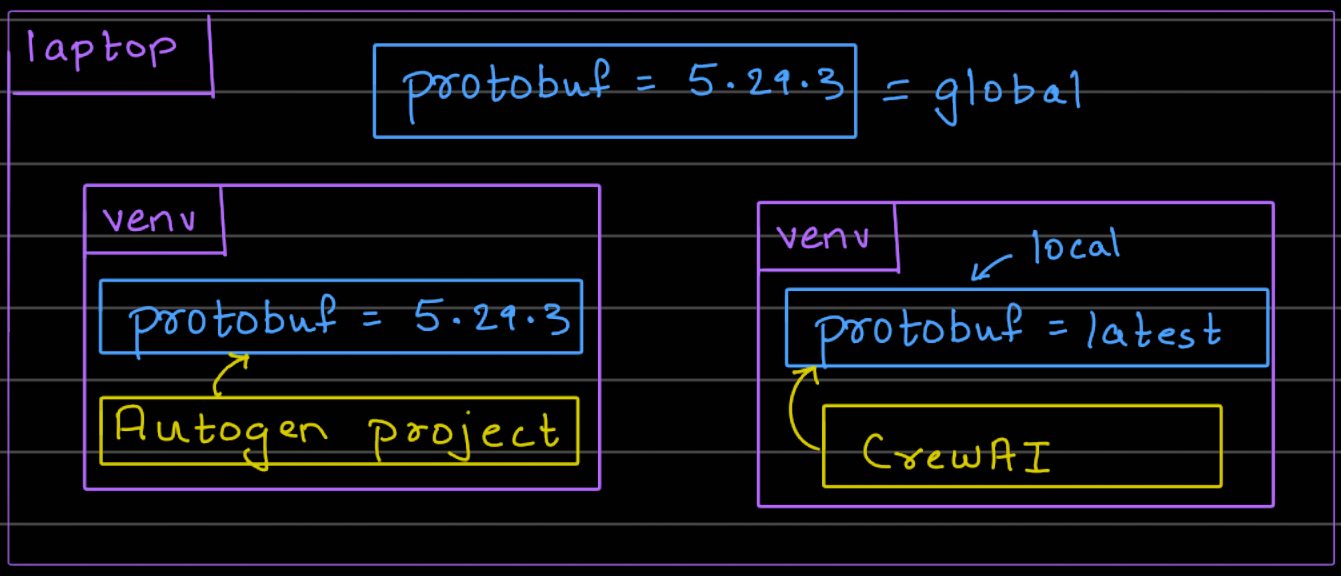
Role = Formatter
Task = Format the article in md
Model = GPT OSS 20b

↓

Role = Reviewer
Task = Review the article
Model = GPT OSS 120b

4 Agents


Project
Laptop
main.py
llm.py
pinecone.py
google_sheet.py
linkedin.py
.env
Ollama API key
Pinecode API key
Google API key
LinkedIn API key
.gitignore ← .env , .venv
requirements.txt ←
Git
version 1
version 2
← git add
git commit
git push
Github



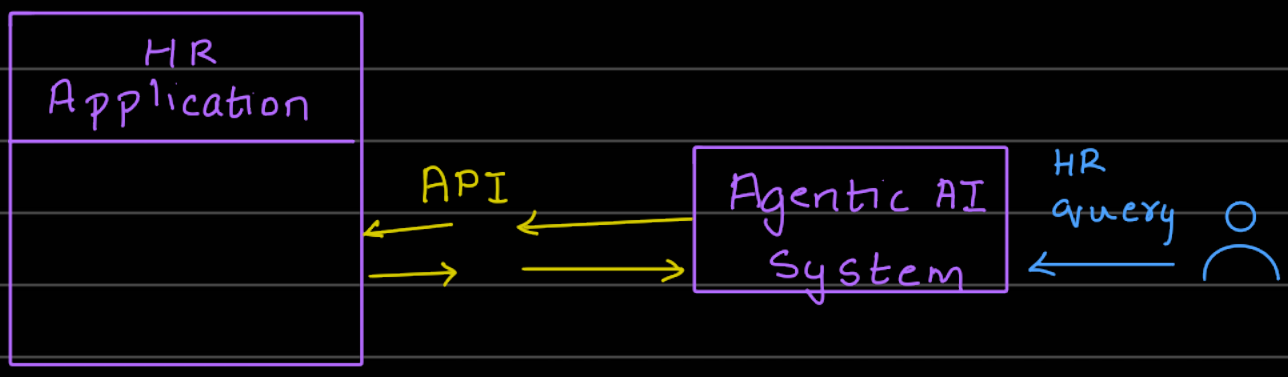
.venv = Virtual environment

MCP → Model Context Protocol

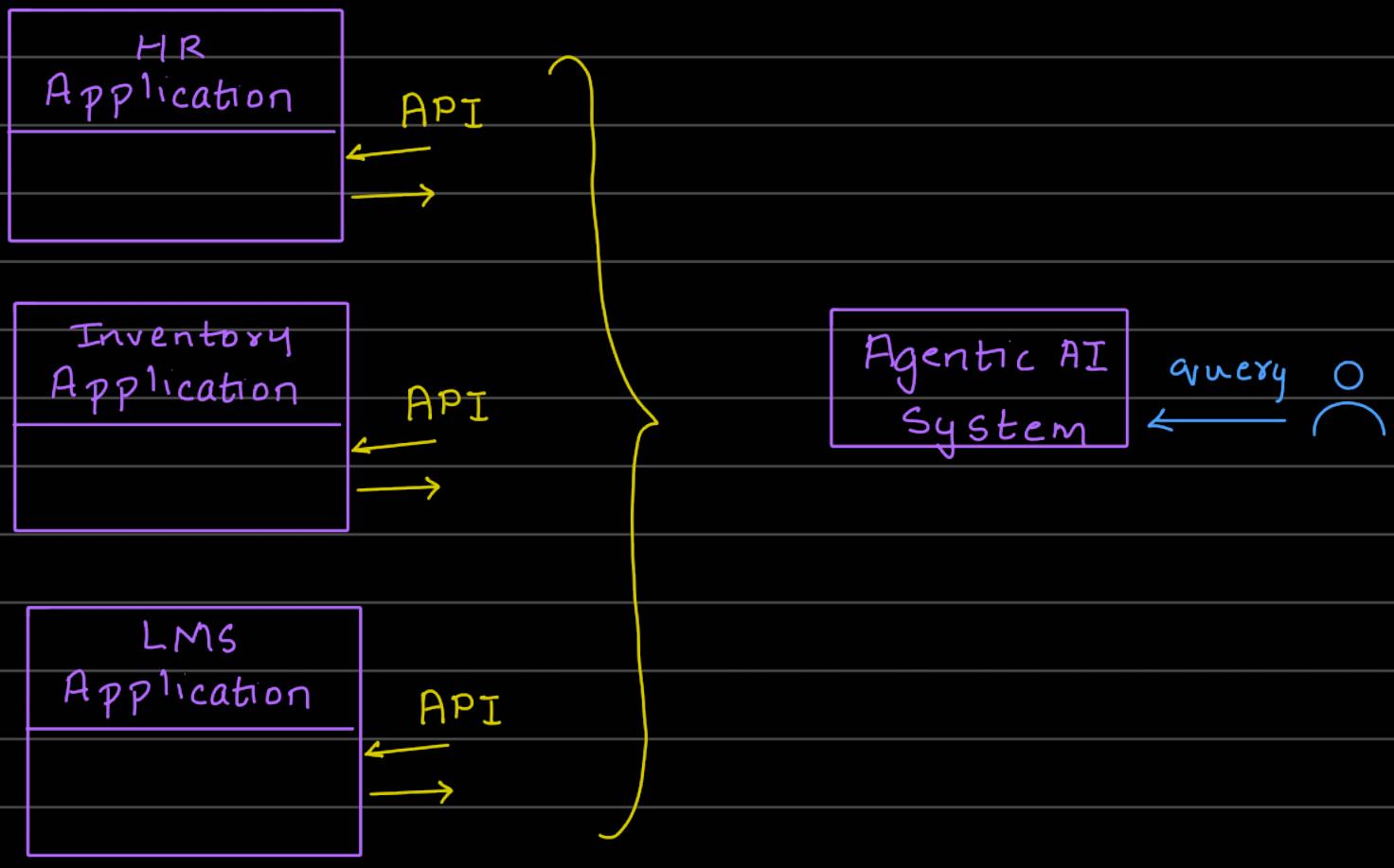
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Set of rules

1) Ownership
2) Scalability
3) Security & Control



LangChain

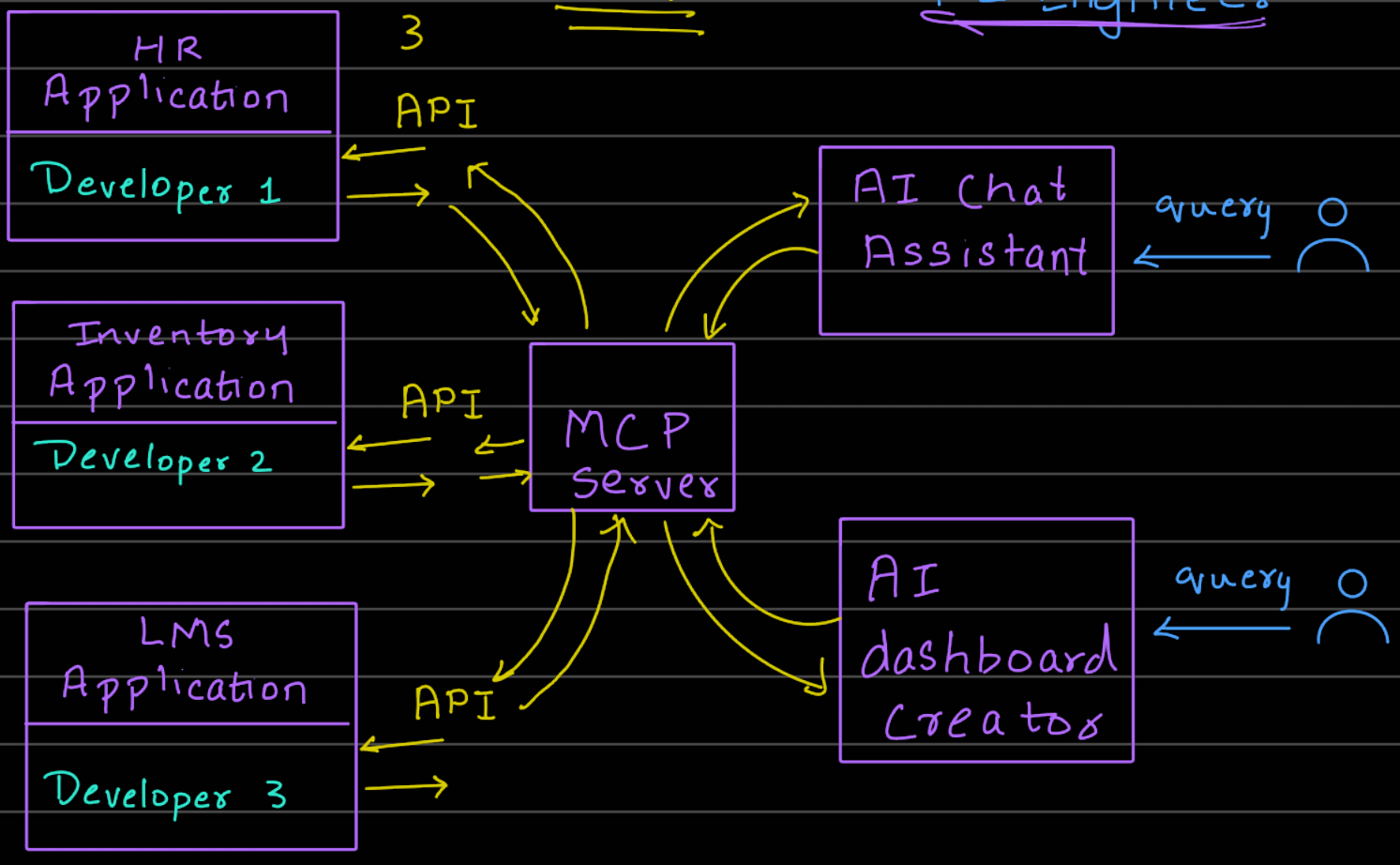


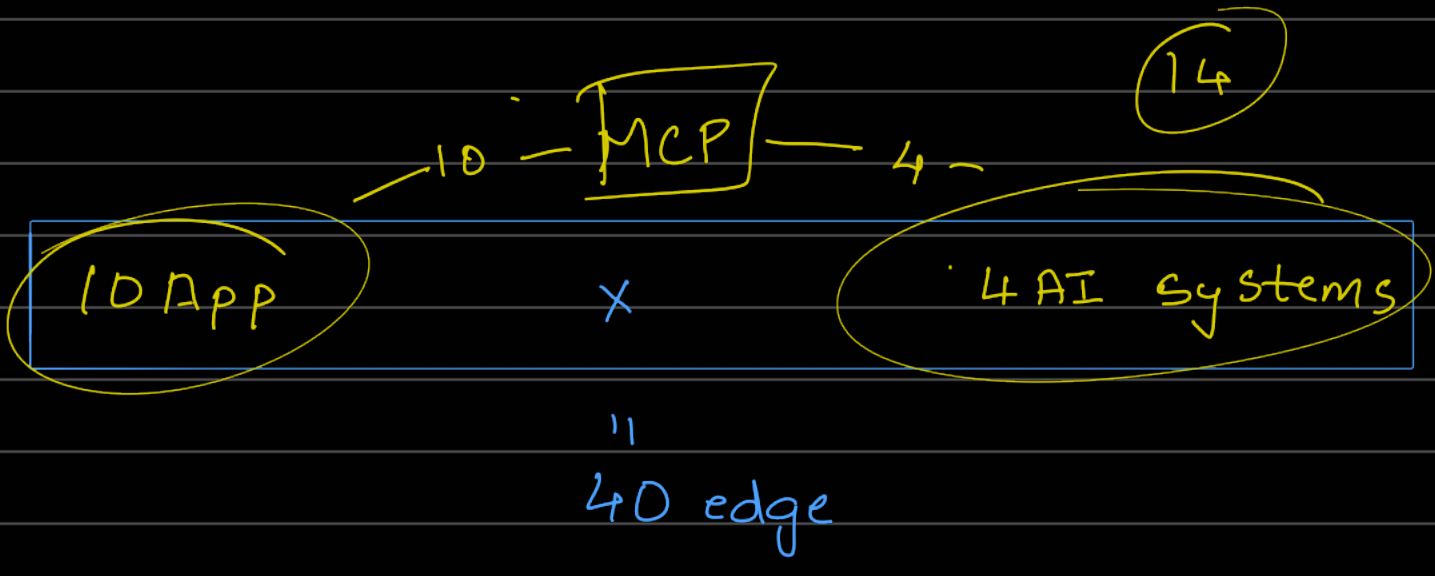
Langgraph

Backend

MCP

AI Engineer





11

40 edge

50 App

50 — MCP — 10

10 App



MCP Architecture: Who Does What?

Example: AI Inventory Assistant

Tool execution happens here
The MCP server runs the tool implementation

User
"Check which products are low on stock"

1. Request

MCP Host / AI App
Claude Desktop, ChatGPT-compatible host, VS Code

LLM
Decides which tool to use

2. Selects tool

MCP Client
Discovers tools and sends tool-call requests

3. tools/call request

MCP Server
Exposes tools and executes tool code

4. Runs query / action

Inventory Database / Business System
Real data and business records

Result

Result

6. Result (AI app answers the user)

How It Works (Step by Step)

1. User asks a question
2. LLM chooses a tool
3. MCP Client sends the request
4. MCP Server executes the tool
5. Database/API returns data
6. AI app answers the user

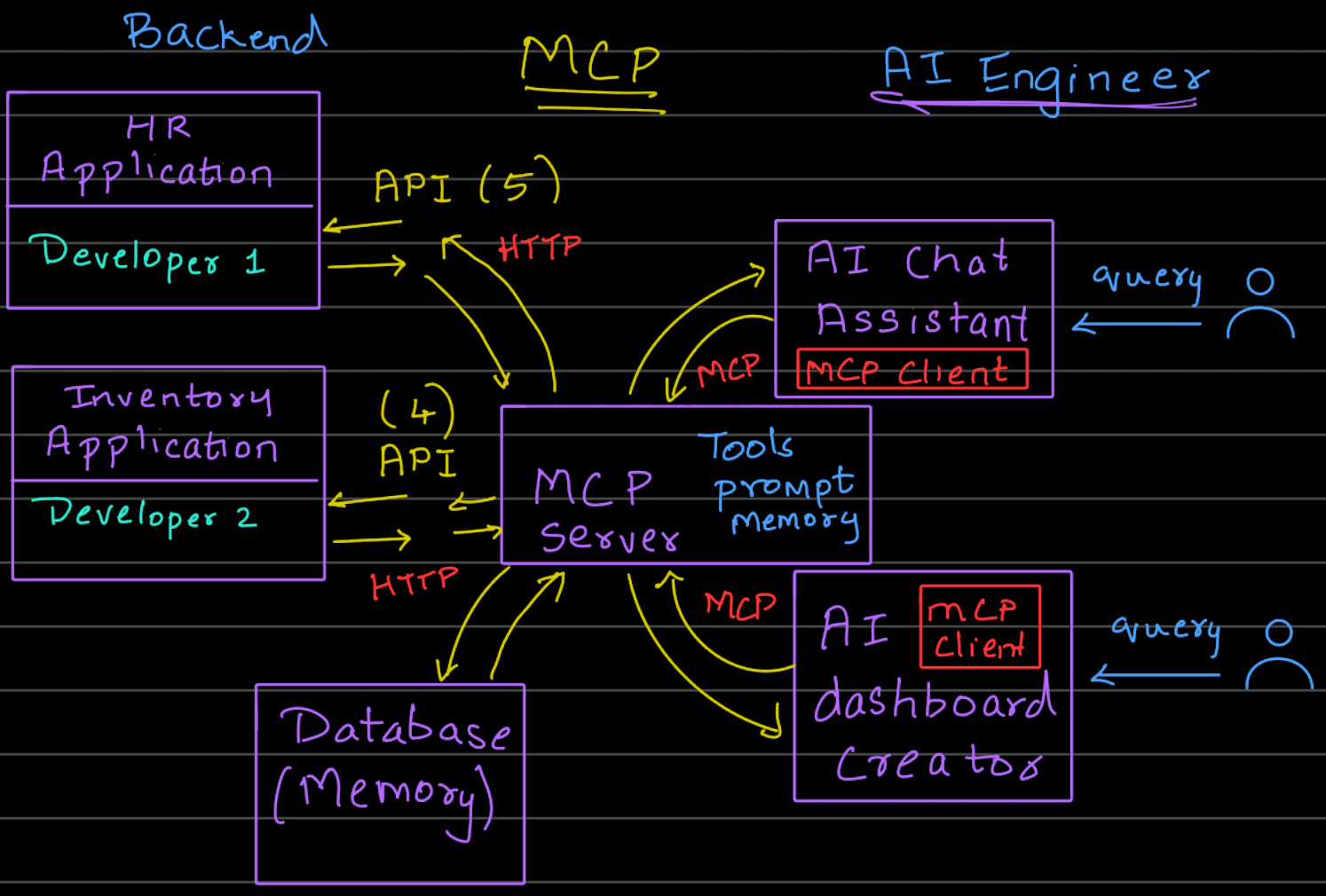
What Each Component Does

User	Asks a lquestion in natural language.
MCP Host / AI App	Runs the LLM and MCP Client. (Claude Desktop, ChatGPT-compatible host, VS Code)
LLM	Decides which tool to use.
MCP Client	Discovers tools and sends tool-call requests.
MCP Server	Exposes tools and executes tool code.
Inventory Database / Business System	Real data and business records.

Key Takeaways

MCP Client
Requests tool execution.

MCP Server
Provides and executes tools.



Real Application

(ReactJS, Angular)

Frontend → HTML CSS Javascript

Frontend → Python (Steamlit) UI

Demo, PoC

Python Backend → FastAPI